

Word Similarity

Thesaurus-based Word Similarity

Hyponymy: "is-a" relationship

Example: car is a kind of vehicle

Meronymy: "part-of" relationship

Example: wheel is part of a car

Troponymy: "manner-of" relationship (used for verbs)

Example: sprint is a manner of run

- We could use anything in the thesaurus:
 - Meronymy, hyponymy, troponymy
 - Glosses and example sentences
- In practice, "thesaurus-based" methods usually use:
 - the is-a/subsumption/hypernymy hierarchy
 - and sometimes the glosses too (short definitions)
- Word similarity vs. word relatedness
 - Similar words are near-synonyms
 - Related words could be related any way
 - ★ car, gasoline : related, but not similar
 - ★ car, bicycle: similar

Gloss and Example

Crucially needed for concept explication, wordnet building using another wordnet and wordnet linking.

{earthquake, **quake**, temblor, seism} -- (shaking and vibration at the surface of the earth resulting from underground movement along a fault plane or from volcanic activity)

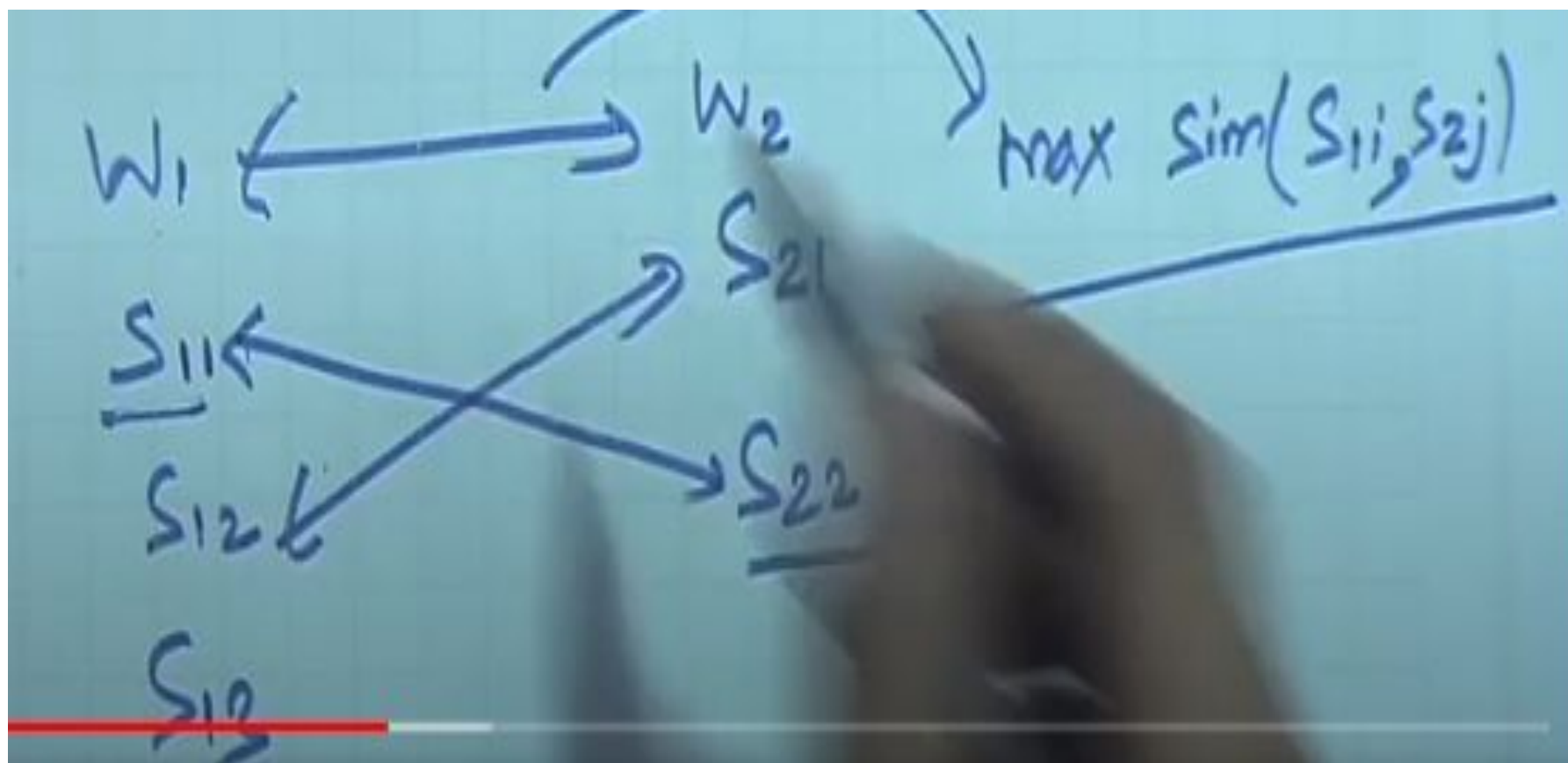
Thesaurus-based Word Similarity

Path-based similarity

Basic Idea

- Two words are similar if they are nearby in the hypernym graph
- $pathlen(c_1, c_2)$ = number of edges in shortest path (in hypernym graph) between senses c_1 and c_2
- $sim_{path}(c_1, c_2) = \frac{1}{1 + pathlen(c_1, c_2)}$

Give the relation between two senses

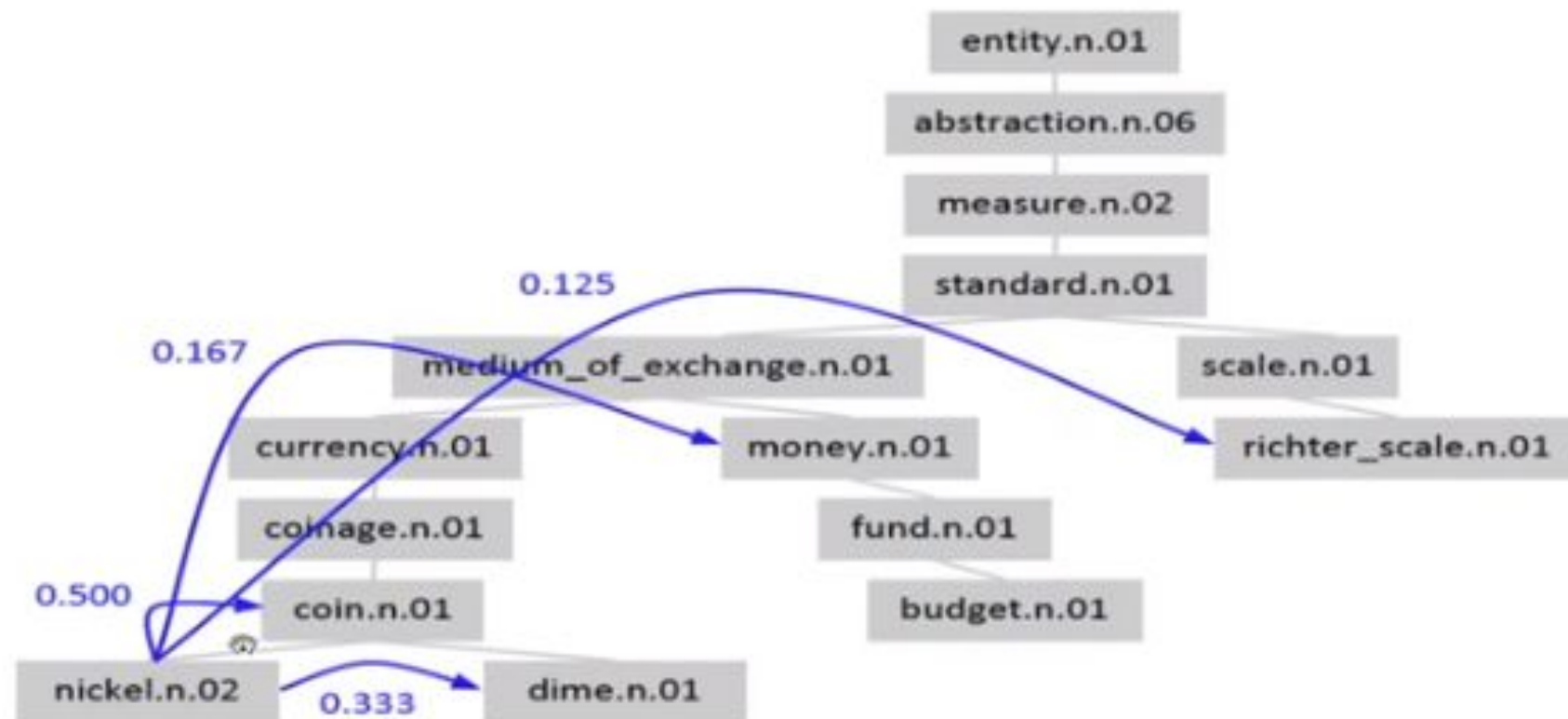


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- $sim(\hat{w}_1, w_2) = \max_{c_1 \in senses(w_1), c_2 \in senses(w_2)} sim(c_1, c_2)$

Shortest path in the hierarchy



Leacock-Chodorow (L-C) Similarity

L-C similarity

$$\text{sim}_{LC}(c_1, c_2) = -\log(\text{pathlen}(c_1, c_2)/2d)$$

d : maximum depth of the hierarchy

Problems with L-C similarity

- Assumes each edge represents a uniform distance
- 'nickel-money' seems closer than 'nickel-standard'
- We want a metric which lets us assign different "lengths" to different edges - but how?

Concept probability models

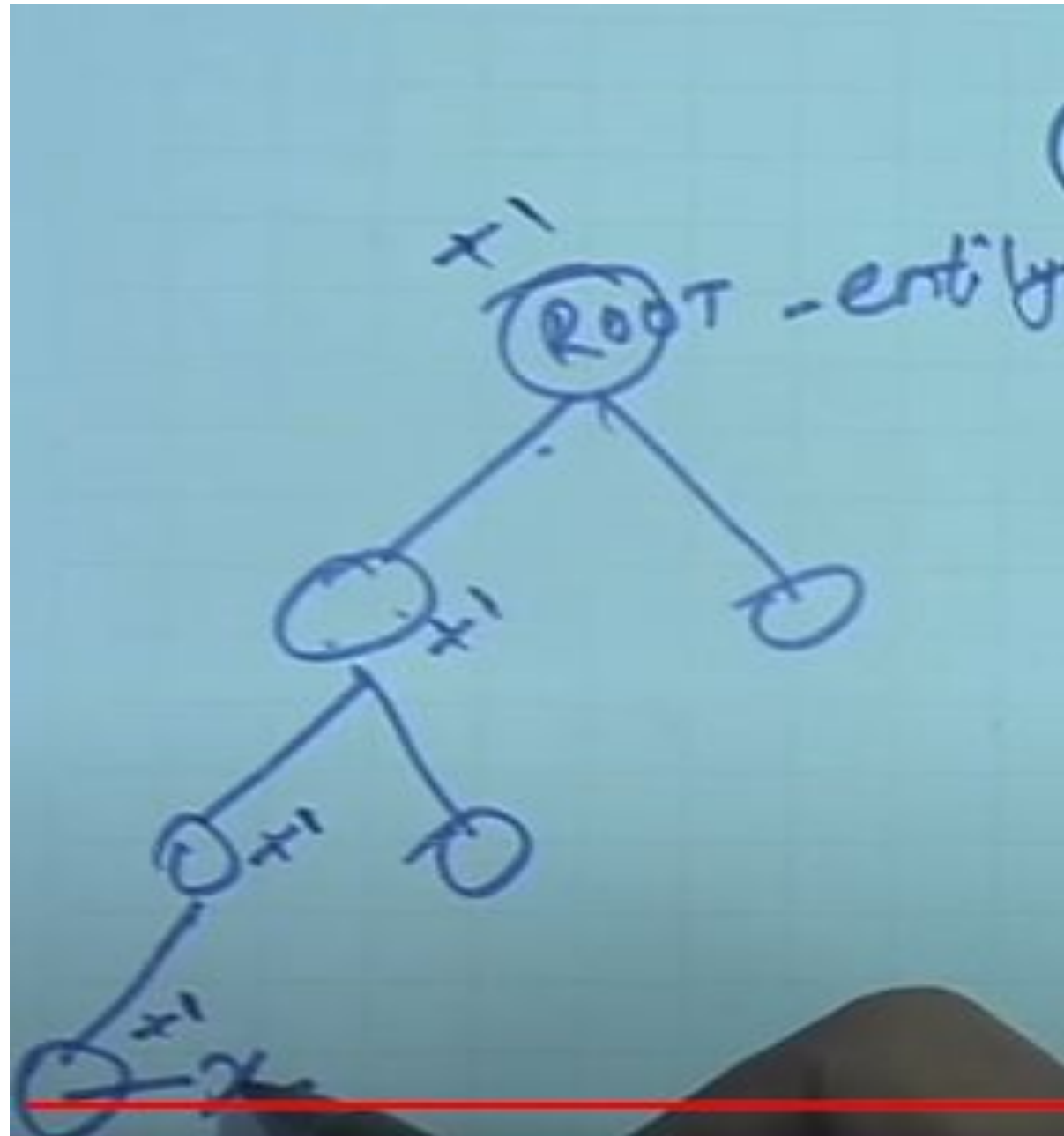
This framework assigns a probability to each concept (synset) in a taxonomy like WordNet, reflecting how often that concept (or any of its hyponyms) appears in real text. It turns the semantic hierarchy into a probabilistic model grounded in corpus evidence.

Concept probabilities

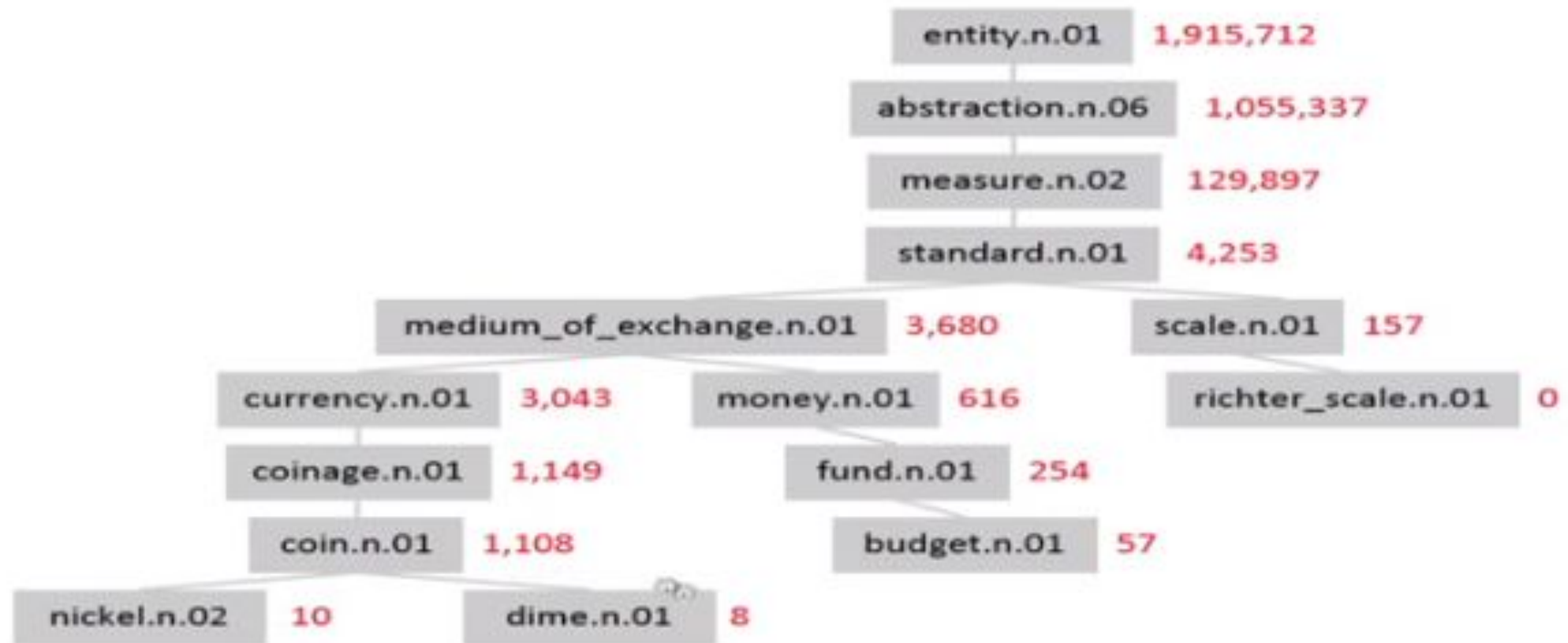
- For each concept (synset) c , let $P(c)$ be the probability that a randomly selected word in a corpus is an instance (hyponym) of c
- $P(ROOT) = 1$
- The lower a node in the hierarchy, the lower its probability

Estimating concept probabilities

- Train by counting “concept activations” in a corpus
- Each occurrence of *dime* also increments counts for *coin*, *currency*, *standard*, etc.



Example : concept count



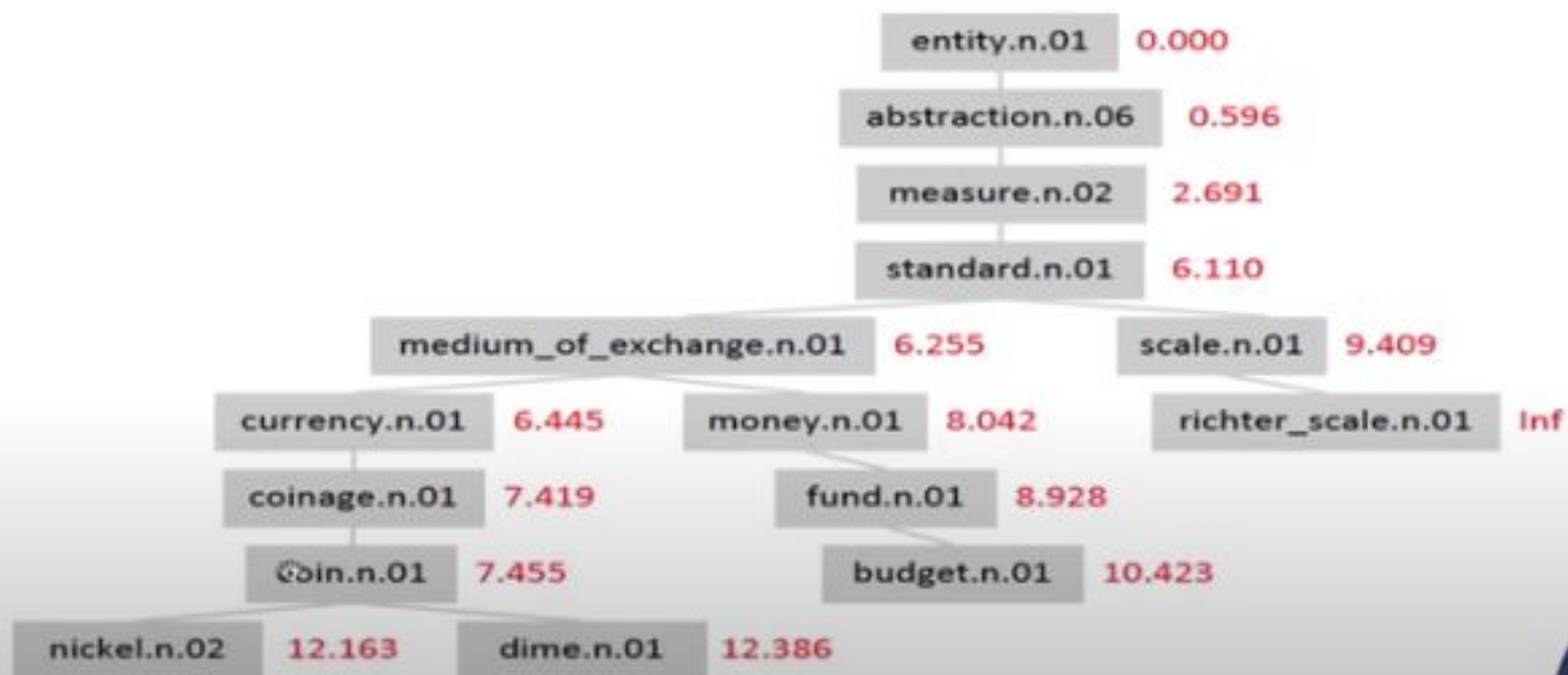
Information content

IC quantifies how “specific” or “informative” a concept is.

Information content

- Information content: $IC(c) = -\log P(c)$
- Lowest common subsumer : $LCS(c_1, c_2)$: the lowest node in the hierarchy that subsumes (is a hypernym of) both c_1 and c_2
- We are now ready to see how to use information content (IC) as a similarity metric.

Example : Information content



Resnik Similarity

Resnik Similarity

- Intuition: how similar two words are depends on how much they have in common
- It measures the commonality by the information content of the lowest common subsumer
- $sim_{resnik}(c_1, c_2) = IC(LCS(c_1, c_2)) = -\log P(LCS(c_1, c_2))$